MinImageScorer: Error-Driven Difficulty Scoring for Guided Copy-Paste Augmentation in Agricultural Object Detection

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***Abstract*—Copy-paste augmentation is widely applied in object detection but is rarely evaluated against the specific failure mode a dataset exhibits. We propose MinImageScorer, an error-driven framework that scores per-object detection difficulty from baseline model failures, combining confidence and spatial accuracy penalties calibrated via Pearson correlation optimization. We validate the score on two agricultural benchmarks—MinneApple and Global Wheat Head (GWHD)—demonstrating that it reliably ranks objects and images by difficulty (Pearson r up to 0.82 with miss rate) and correctly recovers domain- and size-level failure patterns without supervision. We then use the score to guide copy-paste augmentation in a controlled three-seed study. Score-guided selection outperforms random selection on MinneApple (ΔAP=$+0.013$) but neither condition recovers the no-augmentation baseline. Feature-space analysis confirms that augmented images shift the training distribution away from the test set, explaining the degradation. These findings establish that MinImageScorer correctly identifies hard objects, and that copy-paste utility depends on the dataset's underlying failure mode rather than on selection strategy alone.**

***Index Terms*—object detection, copy-paste augmentation, difficulty scoring, agricultural detection, small objects, domain shift**

I. Introduction

Agricultural object detection benchmarks such as MinneApple [**?**] and Global Wheat Head (GWHD) [**?**] present distinct structural failure modes. MinneApple is dominated by high-density small objects near the resolution floor of standard backbones, while GWHD exhibits substantial cross-domain visual variation across geographic sources. Copy-paste augmentation has been shown to improve small-object AP on MS COCO [**?**], [**?**], but those datasets lack structural resolution limits or domain shift—the conditions that dominate agricultural benchmarks.

A critical gap remains: practitioners apply copy-paste without first diagnosing whether the dataset's failure mode is addressable by adding more instances. When failure is structural, copy-paste may corrupt the training distribution rather than improve it. No prior work provides an empirical scoring framework to diagnose this mismatch. We make three contributions:

1) **MinImageScorer**: a per-object difficulty score derived from model failures, with data-driven weight calibration requiring no domain assumptions.
2) **Empirical score validation**: the score reliably correlates with miss rate and false-positive rate, and correctly recovers domain-level and object-size-level difficulty rankings without supervision.
3) **Controlled augmentation study**: score-guided copy-paste is compared to random selection across three seeds, with feature-space analysis explaining the outcome.

II. Methodology

A. MinImageScorer Formulation

Let $G(I)$ be the ground-truth objects in image I and $\mathcal{P}(g;\tau)$ the model predictions satisfying $\mathrm{IoU}(p,g) \geq \tau$. The *object difficulty score* is:

$$S_{\mathrm{obj}}(g) = \begin{cases} \min\limits_{p \in \mathcal{P}(g;\tau)} \big[\alpha\,(1-\mathrm{conf}(p)) + \beta\,(1-\mathrm{IoU}(p,g))\big], & \mathcal{P}(g;\tau) \neq \\ \alpha+\beta, & \text{otherwise.} \end{cases} \quad (1)$$

Missed detections receive the maximum penalty $\alpha{+}\beta$. The *image-level score* aggregates object difficulty and penalizes false positives:

$$S_{\mathrm{img}}(I) = w_1 \cdot \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\mathrm{obj}}(g) \;+\; w_2 \cdot \mathrm{FPrate}(I). \quad (2)$$

We use two prediction sets: *low-confidence* predictions (dense, τ=0.01) for S_{obj}, capturing near-misses and weak detections; and *optimal-threshold* predictions for FPrate, reflecting deployment-level errors. Weights α, β, w_1, w_2 are calibrated to maximize Pearson correlation with per-image miss rate and FP rate on the training set.

B. Score-Guided Copy-Paste Pipeline

(1) Train a baseline YOLOv8s [**?**] with all built-in augmentations disabled. (2) Run inference to collect confidence and IoU signals. (3) Compute S_{obj} and S_{img}. (4) For score-guided augmentation, sample objects proportionally to S_{obj} via exponential weighting; for random, sample uniformly. (5) Paste 2–8 objects per augmented image (within-image only) and retrain. Both conditions are identical except for the selection policy.

FIGURE 1 PLACEHOLDER
Scatter: S_{img} vs. per-image miss rate
Left: MinneApple Right: GWHD
Each point = one training image; regression line shown.

Fig. 1. Score vs. per-image miss rate on training images. MinneApple (Pearson r=0.70); GWHD (r=0.82, w/ trad. aug). Higher score correctly predicts harder images on both datasets.

TABLE I
PEARSON CORRELATION OF S_{img} WITH PER-IMAGE MISS RATE AND FP RATE (TRAINING SET, W/ TRADITIONAL AUGMENTATION).

Dataset	r(score, miss rate)	r(score, FP rate)	Inter-seed r (mean)
MinneApple	0.698	0.688	0.806
GWHD	0.816	0.818	0.900

III. Score Validation

Before applying the score to guide augmentation, we validate that it reliably measures detection difficulty across two structural axes: per-image failure correlation and per-domain/object-size rankings.

A. Correlation with Model Failures

Table **??** reports Pearson correlation between S_{img} and per-image miss rate / FP rate, computed on training images across both datasets. All cases show strong positive correlation. The strongest relationship is on GWHD with traditional augmentation (r=0.82 vs. miss rate, r=0.82 vs. FP rate), where the baseline model is stable and the score signal is clean. On MinneApple without traditional augmentation, stability drops (mean inter-seed Pearson $\approx$0.67), because unstabilized models produce fluctuating confidence scores on densely clustered small objects. This confirms that score reliability depends on baseline model stability, not on the score design itself.

B. Domain- and Size-Level Validity

Beyond per-image correlation, we verify that the score recovers structural difficulty known from ground-truth AP. On GWHD, Table **??** shows that domains with lower AP receive higher mean scores: the *arvalis 2* domain (AP$=$ 0.365) receives a mean score of 0.431, while the easiest domain *rres 1* (AP$=$ 0.572) scores only 0.260. This domain ranking is recovered *without access to test labels*—purely from training-set model failures.

On MinneApple, small objects ($\leq$32px) average a score of 0.378 vs. 0.222 for medium objects, perfectly matching the AP gap (0.132 vs. 0.522). Together, these results confirm that MinImageScorer captures both image-level and structural difficulty without domain-specific tuning.

TABLE II
GWHD DOMAIN AP VS. MEAN S_{img}. DOMAINS WITH LOWER AP CONSISTENTLY RECEIVE HIGHER SCORES (NO TEST LABELS USED).

Domain	n img	AP	AP_{50}	Mean Score
rres 1	363	0.572	0.962	0.260
inrae 1	146	0.523	0.954	0.284
ethz 1	592	0.550	0.940	0.291
arvalis 3	457	0.466	0.918	0.347
arvalis 1	827	0.403	0.890	0.369
usask 1	153	0.408	0.874	0.384
arvalis 2	162	0.365	0.813	0.431

TABLE III
THREE-SEED MEAN COCO AP. **BOLD** = BEST COPY-PASTE CONDITION PER DATASET.

Dataset	Condition	AP	AP_{50}	AP_{75}	AP_S	AP_M
MinneApple	Baseline	0.439	0.771	0.452	0.300	0.592
	Random CP	0.415	0.741	0.427	0.279	0.563
	Score CP	**0.428**	**0.759**	**0.440**	**0.286**	**0.581**
GWHD	Baseline	0.507	0.922	0.491	0.136	0.501
	Random CP	0.486	0.905	0.456	0.118	0.480
	Score CP	**0.484**	**0.905**	**0.455**	**0.098**	**0.477**

IV. Augmentation Experiments

A. Setup

All models use YOLOv8s initialized from COCO pretrained weights. Training: MinneApple 200 epochs / patience 10; GWHD 100 epochs / patience 8; batch 16; input 1024$\times$1024; lr 10^{-2}. All Ultralytics built-in augmentations are disabled to isolate copy-paste effects. Augmentation ratio 0.3; scale jitter $[0.9, 1.1]$ (MA) / $[0.8, 1.2]$ (GWHD). Results are 3-seed means.

B. Results and Feature-Space Analysis

Table **??** shows that on MinneApple, score-guided copy-paste outperforms random copy-paste (ΔAP$=+0.013$) across all sub-metrics, but both conditions fall below baseline (-0.024 for Score CP). The AP_{75} drop (0.452$\rightarrow$0.440) indicates degraded localization precision, consistent with hard objects pasted without blending or masks, causing the model to learn edge artifacts. The AP_S drop (0.300$\rightarrow$0.286) reflects that score-selected hard objects are predominantly sub-resolution—they cannot be made more learnable regardless of paste frequency. On GWHD, score-guided and random are indistinguishable (-0.002), and both degrade below baseline (-0.023). Score-guided sampling concentrates objects from domain-minority sources (*arvalis*), amplifying existing domain imbalance.

V. Discussion

Score validity vs. augmentation utility are separable claims. The validation results (Section III) establish that MinImageScorer correctly identifies hard objects and images: domain difficulty rankings are recovered without test labels,

FIGURE 2 PLACEHOLDER
UMAP of YOLOv8s P4 backbone features.
Left: MinneApple Right: GWHD
Groups: original train, test, random CP, score CP.

Fig. 2. Image-level backbone feature projections (YOLOv8s P4, UMAP). Augmented images are displaced from the test distribution relative to original training images on both datasets. On GWHD, Score CP shows greater displacement than Random CP, consistent with domain-minority object concentration.

object size difficulty tracks ground-truth AP, and per-image correlations with miss rate reach $r{=}0.82$ on GWHD. This is the primary contribution.

Why copy-paste fails here. Kisantal *et al.* [**?**] and Ghiasi *et al.* [**?**] demonstrate copy-paste gains on MS COCO, where hard objects are above the resolution floor and within-distribution. Neither condition holds on MinneApple (sub-resolution hard objects) or GWHD (cross-domain images). The score selects the right objects; the method cannot recover from a structural limitation that precedes selection.

Practical implication. Run MinImageScorer before applying copy-paste: if high-score objects are predominantly sub-resolution or cross-domain, copy-paste will degrade performance regardless of selection quality. This constitutes a dataset-level diagnostic criterion absent from prior work.

VI. Conclusion

MinImageScorer reliably quantifies object and image detection difficulty from model failure signals, validated by strong correlation with miss rate and FP rate, and by correctly recovering domain- and size-level difficulty rankings without supervision. Score-guided copy-paste outperforms random selection on MinneApple ($+0.013$ AP) but both strategies degrade below baseline, explained by distributional shift in feature space. On GWHD, domain shift renders copy-paste ineffective under any selection strategy. Future work includes online augmentation to reduce distributional rigidity and domain-constrained pasting for multi-source benchmarks.

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